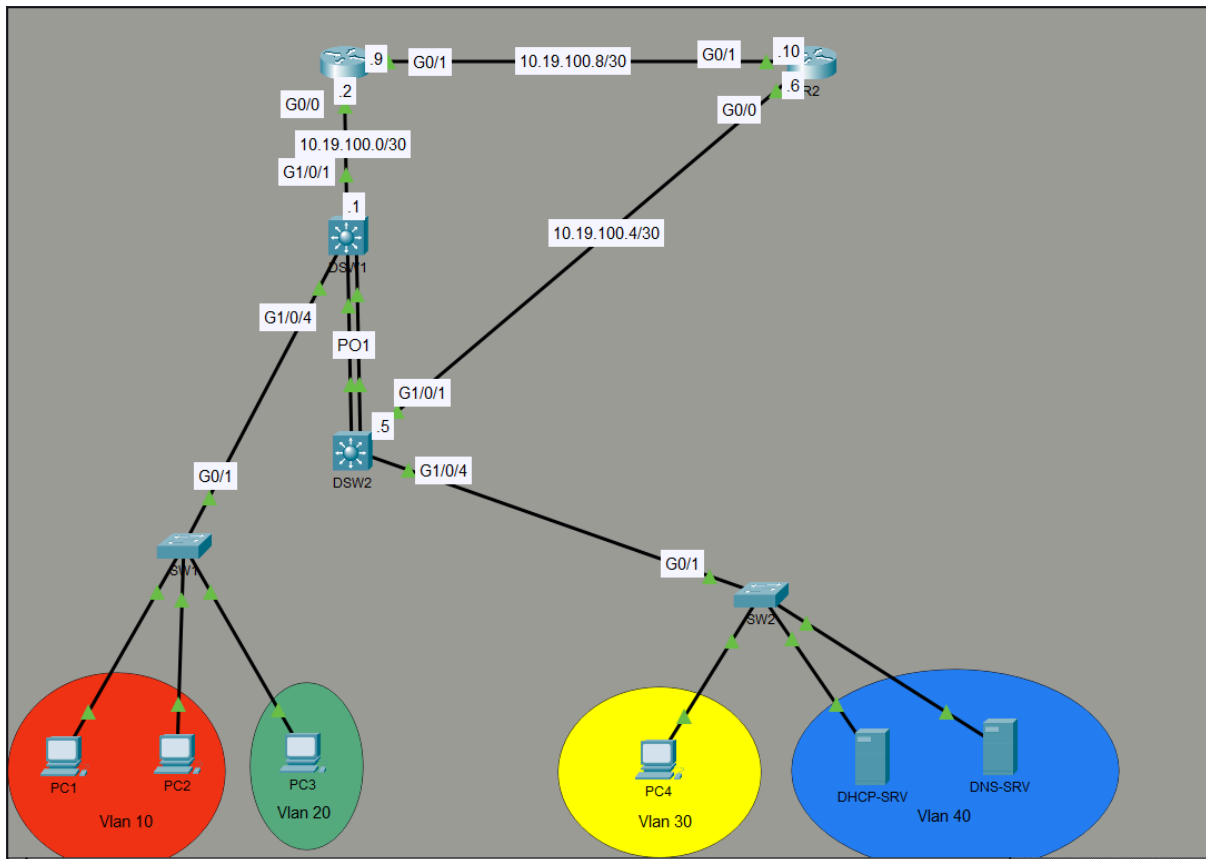




1. Create the Vlans all switches (L2 and L3)

```
SW2(config-vlan)#vlan 10
SW2(config-vlan)#name user
SW2(config-vlan)#vlan 20
SW2(config-vlan)#name it
SW2(config-vlan)#vlan 30
SW2(config-vlan)#name guest
SW2(config-vlan)#vlan 40
SW2(config-vlan)#name servers
SW2(config-vlan)#vlan 99
SW2(config-vlan)#name management
SW2(config-vlan)#do sh vlan br
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
10	user	active	
20	it	active	
30	guest	active	Fa0/1
40	servers	active	Fa0/2, Fa0/3
99	management	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW2(config-vlan)#
```

IP Addressing

VLAN 10 -USERS	VLAN 20 - IT:	VLAN 30 - GUEST:	VLAN 40 - SERVERS:	VLAN 99 - MNGMT:
10.19.10.0/24 HSRP VIP: 10.19.10.1 DSW1: 10.19.10.2 DSW2: 10.19.10.3	10.19.20.0/24 HSRP VIP: 10.19.20.1 DSW1: 10.19.20.2 DSW2: 10.19.20.3	10.19.30.0/24 HSRP VIP: 10.19.30.1 DSW1: 10.19.30.2 DSW2: 10.19.30.3	10.19.40.0/24 HSRP VIP: 10.19.40.1 DSW1: 10.19.40.2 DSW2: 10.19.40.3	10.19.99.0/24 HSRP VIP: 10.19.99.1 DSW1: 10.19.99.2 DSW2: 10.19.99.3

- On the L3 switches, we create the interface Vlans and assign the ip addresses to them.

```
DSW1(config)#int vlan 10
DSW1(config-if)#ip add 10.19.10.2 255.255.255.0
DSW1(config-if)#
```

```
Vlan1          unassigned      YES unset  administratively down down
Vlan10         10.19.10.2        YES manual up          up
Vlan20         10.19.20.2        YES manual up          up
Vlan30         10.19.30.2        YES manual up          up
Vlan40         10.19.40.2        YES manual up          up
Vlan99         10.19.99.2        YES manual up          up
DSW1#
```

```
Vlan1          unassigned      YES unset  administratively down down
Vlan10         10.19.10.3        YES manual up          up
Vlan20         10.19.20.3        YES manual up          up
Vlan30         10.19.30.3        YES manual up          up
Vlan40         10.19.40.3        YES manual up          up
Vlan99         10.19.99.3        YES manual up          up
DSW2#
DSW2#
```

Routed Links:

DSW1 → R1	DSW2 → R2	R1 → R2
10.19.100.0/30	10.19.100.4/30	10.19.100.8/30

DSW1 = 10.19.100.1 R1 = 10.19.100.2	DSW2 = 10.19.100.5 R2 = 10.19.100.6	R1 = 10.19.100.9 R2 = 10.19.100.10
--	--	---------------------------------------

- access the interfaces and assigned the ip address

```
R1(config-if)#int g0/0
R1(config-if)#ip add 10.19.100.2 255.255.255.252
R1(config-if)#no sh
R1(config-if)#
```

```
R1#sh ip int br
Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0 10.19.100.2     YES manual up          up
GigabitEthernet0/1 10.19.100.9     YES manual up          up
GigabitEthernet0/2 unassigned      YES unset   administratively down down
Vlan1              unassigned      YES unset   administratively down down
R1#
```

HSRP

Configure HSRP on VLANs 10, 20, 30, 40 and 99.

DSW1 preferred	DSW2 preferred
VLAN 10 VLAN 40 VLAN 99	VLAN 20 VLAN 30

- On DSW1 we configure the priority higher for vlan 10,40 and 99. and on DSW2 higher for Vlan 20 and 30.

```
interface Vlan10
 mac-address 0001.c739.4501
 ip address 10.19.10.2 255.255.255.0
 standby 10 ip 10.19.10.1
 standby 10 priority 110
 standby 10 preempt
!
interface Vlan20
 mac-address 0001.c739.4502
 ip address 10.19.20.2 255.255.255.0
 standby 20 ip 10.19.20.1
!
interface Vlan30
 mac-address 0001.c739.4503
 ip address 10.19.30.2 255.255.255.0
 standby 30 ip 10.19.30.1
!
interface Vlan40
```

```
DSW2(config-if)#no stand 10 ip 10.19.20.1
DSW2(config-if)#int vlan 20
DSW2(config-if)#standby 20 ip 10.19.20.1
DSW2(config-if)#standby 20 prio 110
DSW2(config-if)#standby 20 pre
DSW2(config-if)#standby 20 preempt
DSW2(config-if)#int vlan 30
DSW2(config-if)#stand
%HSRP-6-STATECHANGE: Vlan20 Grp 20 state Speak -> Standby
%HSRP-6-STATECHANGE: Vlan20 Grp 20 state Standby -> Active
% Incomplete command.
DSW2(config-if)#stand 30 ip 10.19.30.1
DSW2(config-if)#stand 30 prio
DSW2(config-if)#stand 30 priority 110
DSW2(config-if)#stand 30 pre
DSW2(config-if)#stand 30 preempt
```


EtherChannel

Between DSW1 and DSW2:

5. Create a port-channel

```
DSW1(config)#int ra g1/0/2-3
DSW1(config-if-range)#channel-gr
DSW1(config-if-range)#channel-group 1 mo ac
%EC-5-L3DONTBNDL2: Gig1/0/2 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig1/0/3 suspended: LACP currently not enabled on the remote port.
DSW1(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to up
%LINK-5-CHANGED: Interface Port-channel1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up
```

6. Configure it as a trunk and allow vlan 10,20,30,40,99

```
DSW1(config-if-range)#int po1
DSW1(config-if)#sw mo tru
DSW1(config-if)#sw mo trunk

DSW1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up
%LINK-3-UPDOWN: Interface Port-channel1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to up
%LINK-5-CHANGED: Interface Port-channel1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

DSW1(config-if)#
DSW1(config-if)#
DSW1(config-if)#sw tr ll
DSW1(config-if)#sw tr al
DSW1(config-if)#sw tr allowed vlan 10,20,30,
Command rejected: Bad VLAN list
DSW1(config-if)#sw tr allowed vlan 10,20,30,40,99
DSW1(config-if)#
```

```

V199          99      100      Standby  10.19.99.2      local      10.19.99.1
DSW2(config-if)#do sh stand br
                P indicates configured to preempt.
                |
Interface      Grp   Pri  P State      Active          Standby          Virtual IP
Vl10           10    100      Standby  10.19.10.2      local            10.19.10.1
Vl20           20    110 P Active   local           10.19.20.2      10.19.20.1
Vl30           30    110 P Active   local           10.19.30.2      10.19.30.1
Vl40           40    100      Standby  10.19.40.2      local            10.19.40.1
Vl99           99    100      Standby  10.19.99.2      local            10.19.99.1
DSW2(config-if)#

```

OSPF

7. Establish OSPF adjacencies between:

DSW1 ↔ R1

DSW2 ↔ R2

R1 ↔ R2

```

DSW1(config)#router ospf 10
DSW1(config-router)#net 10.19.10.0 0.0.0.255 area 0
DSW1(config-router)#net 10.19.20.0 0.0.0.255 area 0
DSW1(config-router)#net 10.19.100.0 0.0.0.3 area 0

```

```

Neighbor ID      Pri   State             Dead Time   Address        Interface
10.19.100.9       1    FULL/BDR          00:00:34    10.19.100.2    GigabitEthernet1/0/1
DSW1(config-router)#

```

```

R1(config)#router ospf 10
R1(config-router)#net 10.19.100.0 0.0.0.3 area 0
R1(config-router)#net 10.19.100.8 0.0.0.3 area 0
R1(config-router)#

```

```

R1(config-router)#do sh ip osp ne
Neighbor ID      Pri   State             Dead Time   Address        Interface
10.19.100.10      1    FULL/BDR          00:00:36    10.19.100.10   GigabitEthernet0/1
10.19.100.1       1    FULL/DR           00:00:31    10.19.100.1    GigabitEthernet0/0
R1(config-router)#

```

```

R1(config-router)#do show ip route ospf
10.0.0.0/8 is variably subnetted, 9 subnets, 3 masks
O        10.19.10.0 [110/2] via 10.19.100.1, 00:04:25, GigabitEthernet0/0
O        10.19.20.0 [110/2] via 10.19.100.1, 00:04:25, GigabitEthernet0/0
O        10.19.30.0 [110/3] via 10.19.100.10, 00:03:42, GigabitEthernet0/1
O        10.19.40.0 [110/3] via 10.19.100.10, 00:03:42, GigabitEthernet0/1
O        10.19.100.4 [110/2] via 10.19.100.10, 00:03:42, GigabitEthernet0/1

```

DHCP Server Configuration

The DHCP server is located in: VLAN 40 10.19.40.20

But the clients are in: VLAN 10, VLAN 20, VLAN 30

Therefore, the DHCP broadcasts cannot simply cross the Layer 3 boundary.

You must configure the Layer 3 interfaces as **DHCP relay agents**.

DHCP Pools

Create three DHCP pools on **DHCP-SRV**

The screenshot shows the DHCP-SRV configuration interface. On the left is a sidebar with a 'SERVICES' menu containing options like HTTP, DHCP, DHCPv6, TFTP, DNS, SYSLOG, AAA, NTP, EMAIL, FTP, IoT, VM Management, Radius EAP, and PRP. The main area is titled 'DHCP' and shows configuration for the 'FastEthernet0' interface. The 'Service' is set to 'On'. The 'Pool Name' is 'guest'. The 'Default Gateway' is '10.19.30.1'. The 'DNS Server' is '10.19.40.10'. The 'Start IP Address' is '10.19.30.10' and the 'Subnet Mask' is '255.255.255.0'. The 'Maximum Number of Users' is '246'. The 'TFTP Server' is '0.0.0.0' and the 'WLC Address' is '0.0.0.0'. Below this is a table of DHCP pools with columns: Pool Name, Default Gateway, DNS Server, Start IP Address, Subnet Mask, Max User, TFTP Server, and WLC Address. The table contains four rows: 'guest', 'IT', 'users_pool', and 'serverPool'. The first three rows are highlighted with a red border.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
guest	10.19.30.1	10.19.40.10	10.19.30.10	255.255.255.0	246	0.0.0.0	0.0.0.0
IT	10.19.20.1	10.19.40.10	10.19.20.10	255.255.255.0	246	0.0.0.0	0.0.0.0
users_pool	10.19.10.1	10.19.40.10	10.19.10.10	255.255.255.0	246	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	10.19.40.0	255.255.255.0	512	0.0.0.0	0.0.0.0

DHCP Relay

This is the key configuration.

On the **Layer 3 gateway interfaces** for:

VLAN 10

VLAN 20

VLAN 30

```
Enter configuration commands, one per line.  End with CNTL/Z.
DSW1(config)#int vlan 10
DSW1(config-if)#ip help
DSW1(config-if)#ip helper-address 10.19.40.20
DSW1(config-if)#int vlan 20
DSW1(config-if)#ip helper-address 10.19.40.20
DSW1(config-if)#int vlan 30
DSW1(config-if)#ip helper-address 10.19.40.20
DSW1(config-if)#
```

```
DSW2(config)#int vlan 30
DSW2(config-if)#ip hel
DSW2(config-if)#ip hel
DSW2(config-if)#ip help
DSW2(config-if)#ip helper-address ?
  A.B.C.D  IP destination address
DSW2(config-if)#ip helper-address 10.19.40.20 ?
  <cr>
DSW2(config-if)#ip helper-address 10.19.40.20
DSW2(config-if)#int vlan 10
DSW2(config-if)#ip helper-address 10.19.40.20
DSW2(config-if)#int vlan 20
DSW2(config-if)#ip helper-address 10.19.40.20
DSW2(config-if)#
```

This needs to be done on all switches involved in the HSRP.

DHCP + HSRP

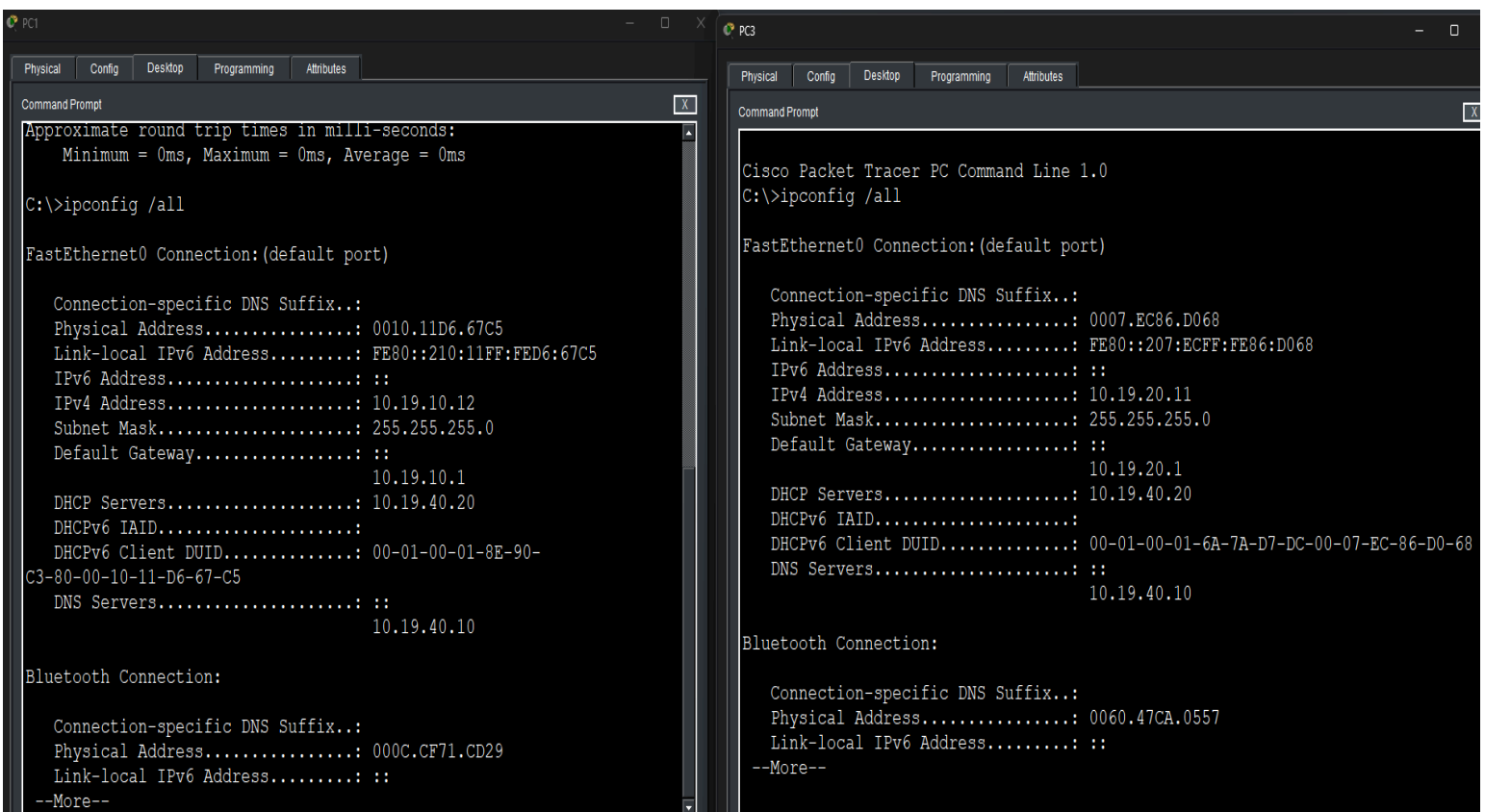
Here's the important enterprise concept.

The client's default gateway is: 10.19.10.1, not 10.19.10.2 neither 10.19.10.3

The client uses the HSRP virtual IP.

This lets DHCP relay continue functioning while HSRP provides gateway redundancy.

Then we set the PCs to received their ip address from the DHCP servers.

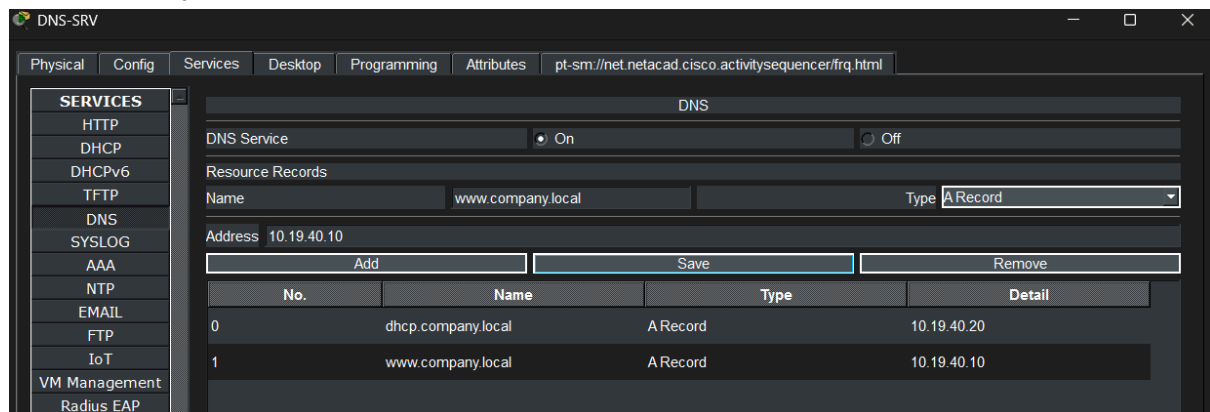


DNS Integration

On **DNS-SRV**, configure:

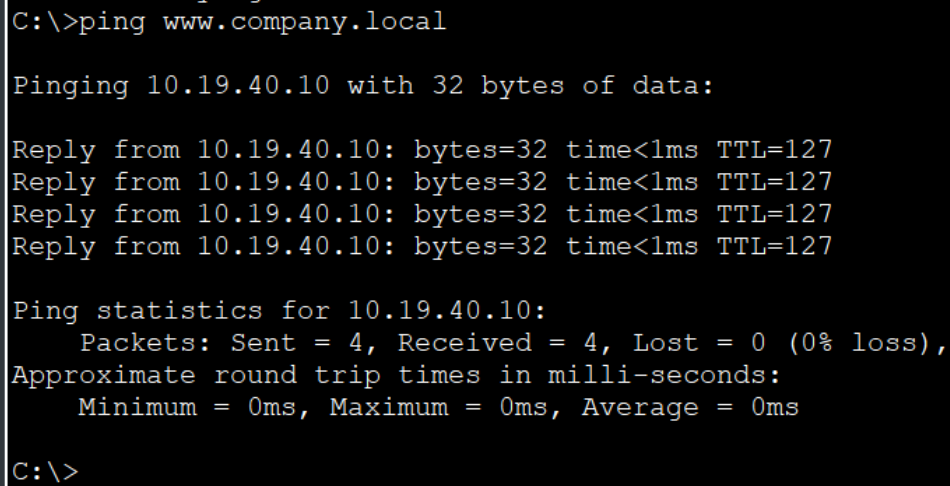
www.company.local → 10.19.40.10

dhcp.company.local → 10.19.40.20



Because DHCP automatically gives clients the DNS server address, you should now be able to test:

ping www.company.local



```
C:\>ping www.company.local

Pinging 10.19.40.10 with 32 bytes of data:

Reply from 10.19.40.10: bytes=32 time<1ms TTL=127
Reply from 10.19.40.10: bytes=32 time<1ms TTL=127
Reply from 10.19.40.10: bytes=32 time<1ms TTL=127
Reply from 10.19.40.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.19.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

ACL Requirement

Guest users must not access internal networks.

Block:

10.19.30.0/24

from reaching:

10.19.10.0/24

10.19.20.0/24

10.19.40.0/24

10.19.99.0/24

But allow Guest to reach external networks.

```

DSW2(config-ext-nacl)#ip access-list extended 110
DSW2(config-ext-nacl)#permit ip 10.19.30.0 0.0.0.255 10.19.100.0 0.0.0.15
DSW2(config-ext-nacl)#int vlan 30
DSW2(config-if)#ip acc
DSW2(config-if)#ip access-group ex
DSW2(config-if)#ip access-group ?
    <1-199>  IP access list (standard or extended)
    WORD      Access-list name
DSW2(config-if)#ip access-group 110 in
DSW2(config-if)#do sh ip acc
Extended IP access list 110
    10 permit ip 10.19.30.0 0.0.0.255 10.19.100.0 0.0.0.15
DSW2(config-if)#

```

```

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: FE80::2E0:8FFF:FEDA:E5A8
IPv6 Address.....: ::
IPv4 Address.....: 10.19.30.11
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        10.19.30.1

```

Bluetooth Connection:

```

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
                        0.0.0.0

```

C:\>ping 10.19.10.10

Pinging 10.19.10.10 with 32 bytes of data:

```

Reply from 10.19.30.3: Destination host unreachable.
Reply from 10.19.30.3: Destination host unreachable.
Reply from 10.19.30.3: Destination host unreachable.
Reply from 10.19.30.3: Destination host unreachable.

```

Ping statistics for 10.19.10.10:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.19.100.1

Pinging 10.19.100.1 with 32 bytes of data:

```

Reply from 10.19.100.1: bytes=32 time<1ms TTL=255
Reply from 10.19.100.1: bytes=32 time<1ms TTL=255
Reply from 10.19.100.1: bytes=32 time<1ms TTL=255

```

